Competing Risks and Multi-State Models

Concepts, methods and software

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Part IV

Interpretation; Summary

Contents:

1. Stacked data set: advantages
2. From hazard regression to cumulative incidence
3. Which quantity to use?

Outline

Combined analysis
- Stacked data set
- Analyses
- Standard error

Advantages combined analysis
- Test for equality
- Assume equality
- Proportional baseline hazards

From regression on hazard to cumulative scale
- Competing risks

Which Type of Analysis to Choose?
- Choices…
- Three types of study questions
- Etiology: marginal or competing risks analysis
- Competing risks: type of quantity

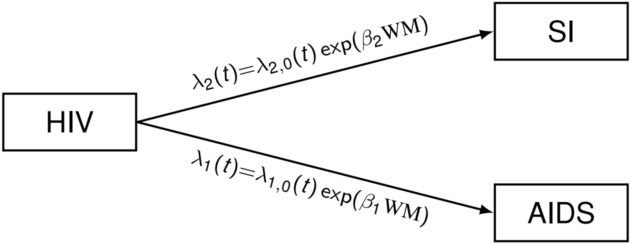
Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number "in risk set" (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \ldots, Z_{iq})$

Example: effect CCR5Δ32 on AIDS and SI



Analysis per event type

We cannot

- test whether effect of covariable differs per event type
- assume equality of effect for some (or all) event types
- test for proportional baseline hazards or assume them proportional

Similar issues as with stratified regression analysis

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Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type

patnr	time	status	status2	cause	failcode	ccr5
14	5.05	0	0	event-free	AIDS	WW
3	2.23	1	1	AIDS	AIDS	WW
15	10.20	1	1	AIDS	AIDS	WM
8	8.61	2	0	SI	AIDS	WW
14	5.05	0	0	event-free	SI	WW
3	2.23	1	0	AIDS	SI	WW
15	10.20	1	0	AIDS	SI	WM
8	8.61	2	1	SI	SI	WW

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Stratified Cox model

Used if baseline hazard may differ per subgroup (nonproportional hazards)

$$h_g(t \mid \mathbf{Z}_i) = h_{g,0}(t) \exp(\beta^\top \mathbf{Z}_i) \ .$$

with g subgroup ("stratum") to which person i belongs.

Stratified Cox model

Used if baseline hazard may differ per subgroup (nonproportional hazards)

$$h_g(t \mid \mathbf{Z}_i) = h_{g,0}(t) \exp(\beta^\top \mathbf{Z}_i) \, .$$

with g subgroup ("stratum") to which person i belongs. Maximise product of partial likelihoods per stratum

$$L(\beta) = \prod_{g \in \mathcal{G}} L^g(\beta)$$

with $L^g(\beta)$ partial likelihood over individuals in stratum g

- If all parameters differ by stratum, same as fitting separate Cox model per stratum
- In R:

```
coxph(Surv(time,status)~strata(var) + covar,
                                      data=...)
```

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Combined analysis I: interaction term

```
coxph(Surv(time, status2) ~ ccr5 * strata(failcode),
                                  data = aidssi.stack)
```

which yields

```
                  coef exp(coef) se(coef)     z       p
ccr5WM          -1.236     0.291    0.307 -4.02 5.7e-05
ccr5WM:strata(fa  0.982     2.669    0.389  2.53 1.2e-02

Likelihood ratio test=23.2  on 2 df, p=9.29e-06  n= 648,
                                   number of events= 220
   (10 observations deleted due to missingness)
```

Compare with −1.236 and −0.254 from analysis per event type.

Analysis per risk type (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
------------------------
         coef exp(coef) se(coef)      z        p
ccr5WM  -1.24     0.291    0.307  -4.02 5.7e-05

Likelihood ratio test=22  on 1 df, p=2.76e-06   n= 324,
                                   number of events= 113
   (5 observations deleted due to missingness)
```

```
coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
------------------------
         coef exp(coef) se(coef)      z        p
ccr5WM -0.254     0.776    0.238 -1.07    0.29

Likelihood ratio test=1.19  on 1 df, p=0.275  n= 324,
                                   number of events= 107
   (5 observations deleted due to missingness)
```

Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type
  and adapt covariables:

patnr	time	status	status2	cause	failcode	ccr5	ccr.comb
14	5.05	0	0	event-free	AIDS	WW	WW
3	2.23	1	1	AIDS	AIDS	WW	WW
15	10.20	1	1	AIDS	AIDS	WM	WM.AIDS
8	8.61	2	0	SI	AIDS	WW	WW
14	5.05	0	0	event-free	SI	WW	WW
3	2.23	1	0	AIDS	SI	WW	WW
15	10.20	1	0	AIDS	SI	WM	WM.SI
8	8.61	2	1	SI	SI	WW	WW

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Combined analysis II: compound variable

```
coxph(Surv(time, status2) ~ ccr.comb + strata(failcode),
                                         data = aidssi.stack)
```

Same effect estimates and p-values as in separate analyses

```
                    coef exp(coef) se(coef)     z       p
ccr.combWM.AIDS   -1.236     0.291    0.307 -4.02 5.7e-05
ccr.combWM.SI     -0.254     0.776    0.238 -1.07 2.9e-01

Likelihood ratio test=23.2  on 2 df, p=9.29e-06  n= 648,
                                   number of events= 220
   (10 observations deleted due to missingness)
```

Analysis per risk type (in R)

```
coxph(Surv(time, status==1) ~ ccr5, data = aidssi)
------------------------
         coef exp(coef) se(coef)      z       p
ccr5WM  -1.24     0.291    0.307  -4.02 5.7e-05

Likelihood ratio test=22  on 1 df, p=2.76e-06   n= 324,
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coxph(Surv(time, status==2) ~ ccr5, data = aidssi)
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Stacked data set for combined analysis on cause-specific hazard

- Combine data sets for each event type
  and adapt covariables:

patnr	time	status	status2	cause	failcode	ccr5	ccr.comb	ccrAIDS	ccrSI
14	5.05	0	0	event-free	AIDS	WW	WW	0	0
3	2.23	1	1	AIDS	AIDS	WW	WW	0	0
15	10.20	1	1	AIDS	AIDS	WM	WM.AIDS	1	0
8	8.61	2	0	SI	AIDS	WW	WW	0	0
14	5.05	0	0	event-free	SI	WW	WW	0	0
3	2.23	1	0	AIDS	SI	WW	WW	0	0
15	10.20	1	0	AIDS	SI	WM	WM.SI	0	1
8	8.61	2	1	SI	SI	WW	WW	0	0

- Baseline hazard differs by event type: stratified Cox model, with `failcode` as stratum

Combined analysis III: type-specific covariables

- The variables `ccrSI` and `ccrAIDS` are examples of type-specific covariables
- Instead of specifying (in separate models k)

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t)\, \exp(\boldsymbol{\beta}_k^\top \mathbf{Z}_i)\,,$$

we can equivalently specify (in one model)

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{k,0}(t)\, \exp(\boldsymbol{\beta}^\top \mathbf{Z}_{k,i})\,,$$

- $\mathbf{Z}_{k,i}$ equals $\mathbf{Z}_i$ for cause k, and has value 0 otherwise

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- $\mathbf{Z}_{k,i}$ equals $\mathbf{Z}_i$ for cause k, and has value 0 otherwise
- Eg. Z_i = CCR5 genotype coding (`WM`=1, `WW`=0); AIDS: $k = 1$; SI: $k = 2$:

$$\texttt{ccrAIDS}_i = Z_{1,i} = \begin{cases} Z_i, & \text{if } \texttt{failcode=AIDS} \\ 0, & \text{if } \texttt{failcode=SI} \end{cases}$$

and

$$\texttt{ccrSI}_i = Z_{2,i} = \begin{cases} 0, & \text{if } \texttt{failcode=AIDS} \\ Z_i, & \text{if } \texttt{failcode=SI} \end{cases}$$

Combined analysis III: results

```
coxph(Surv(time, status2) ~ ccrAIDS + ccrSI +
                     strata(failcode), data = aidssi.stack)
```

Same effect estimates and p-values as before

```
           coef exp(coef) se(coef)     z       p
ccrAIDS  -1.236     0.291    0.307 -4.02 5.7e-05
ccrSI    -0.254     0.776    0.238 -1.07 2.9e-01
```

Why does this work?

Partial likelihood for $\boldsymbol{\beta} = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with `failcode` as stratum: $L(\boldsymbol{\beta}) = L^{\text{AIDS}}(\boldsymbol{\beta}) \times L^{\text{SI}}(\boldsymbol{\beta})$

($R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

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$(R_{\mathrm{AIDS}}(t_i)/R_{\mathrm{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

$$L^{\mathrm{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \mathtt{status2}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \mathtt{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\mathrm{AIDS}}(t_i)} \exp(\beta_1 \times \mathtt{ccrAIDS}_j + \beta_2 \times 0)} \right\} \mathtt{aidssi.stack}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\mathrm{AIDS}}, \beta_{\mathrm{SI}})$ with `failcode` as stratum: $L(\beta) = L^{\mathrm{AIDS}}(\beta) \times L^{\mathrm{SI}}(\beta)$

$(R_{\mathrm{AIDS}}(t_i)/R_{\mathrm{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

$$L^{\mathrm{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\mathrm{AIDS}}(t_i)} \exp(\beta_1 \times \texttt{ccrAIDS}_j + \beta_2 \times 0)} \right\} \quad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \texttt{ccr5}_j)} \right\} \quad \texttt{aidssi}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\text{AIDS}}, \beta_{\text{SI}})$ with `failcode` as stratum: $L(\beta) = L^{\text{AIDS}}(\beta) \times L^{\text{SI}}(\beta)$

($R_{\text{AIDS}}(t_i)/R_{\text{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta_1 \times \texttt{ccrAIDS}_j + \beta_2 \times 0)} \right\} \quad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \texttt{ccr5}_j)} \right\} \quad \texttt{aidssi}$$

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325\ldots \\ \texttt{status2}_i=1}}^{648} \left\{ \frac{\exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_j)} \right\} \quad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=2}}^{324} \left\{ \frac{\exp(\beta_2 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_2 \times \texttt{ccr5}_j)} \right\} \quad \texttt{aidssi}$$

Why does this work?

Partial likelihood for $\beta = (\beta_1, \beta_2) = (\beta_{\mathrm{AIDS}}, \beta_{\mathrm{SI}})$ with `failcode` as stratum: $L(\beta) = L^{\mathrm{AIDS}}(\beta) \times L^{\mathrm{SI}}(\beta)$
($R_{\mathrm{AIDS}}(t_i)/R_{\mathrm{SI}}(t_i)$: risk set under `failcode=AIDS/SI`)

$$L^{\mathrm{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times 0)}{\sum_{j \in R_{\mathrm{AIDS}}(t_i)} \exp(\beta_1 \times \texttt{ccrAIDS}_j + \beta_2 \times 0)} \right\} \quad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left\{ \frac{\exp(\beta_1 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_1 \times \texttt{ccr5}_j)} \right\} \quad \texttt{aidssi}$$

$$L^{\mathrm{SI}}(\beta) = \prod_{\substack{i=325\ldots \\ \texttt{status2}_i=1}}^{648} \left\{ \frac{\exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_i)}{\sum_{j \in R_{\mathrm{SI}}(t_i)} \exp(\beta_1 \times 0 + \beta_2 \times \texttt{ccrSI}_j)} \right\} \quad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=2}}^{324} \left\{ \frac{\exp(\beta_2 \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta_2 \times \texttt{ccr5}_j)} \right\} \quad \texttt{aidssi}$$

- `ccrAIDS` only used for AIDS stratum; `ccrSI` only for SI stratum

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Robust estimate of standard error?

```
coxph(Surv(time, status) ~ ccr.comb + strata(failcode) +
                     cluster(patnr), data = aidssi.stack)
```

gives

```
                    coef   se(coef) robust se      z       p
ccr.combWM.AIDS -1.236      0.307     0.296 -4.18 0.00003
ccr.combWM.SI   -0.254      0.238     0.233 -1.09 0.27000
```

Robust estimate differs from separate analyses

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Advantage I: test whether effects are equal

```
coxph(Surv(time,status) ~ ccr5 + ccrSI + strata(failcode),
                                                aidssi.stack)
```

gives

```
         coef exp(coef) se(coef)     z       p
ccr5WM -1.236     0.291    0.307 -4.02 5.7e-05
ccrSI   0.982     2.669    0.389  2.53 1.2e-02
```

Note: same result via interaction term:

```
coxph(Surv(time, status) ~ ccr5 * strata(failcode),
                                  data = aidssi.stack)
```

Explanation model

patnr	time	status	failcode	ccr5	ccr.comb	ccrAIDS	ccrSI
14	5.05	0	AIDS	WW	WW	0	0
3	2.23	1	AIDS	WW	WW	0	0
15	10.20	1	AIDS	WM	WM.AIDS	1	0
8	8.61	0	AIDS	WW	WW	0	0
14	5.05	0	SI	WW	WW	0	0
3	2.23	0	SI	WW	WW	0	0
15	10.20	0	SI	WM	WM.SI	0	1
8	8.61	1	SI	WW	WW	0	0

$$\lambda_k(t) = \lambda_{k,0}(t) \exp(\beta_1 \times \texttt{ccr5} + \beta_2 \times \texttt{ccrSI}) .$$

AIDS; WW	AIDS; WM	SI; WW	SI; WM
$\lambda_{1,0}(t)$	$\lambda_{1,0}(t)\exp(\beta_1)$	$\lambda_{2,0}(t)$	$\lambda_{2,0}(t)\exp(\beta_1 + \beta_2)$

Outline

Advantage II: assume equality

Same effect of CCR5 mutation on relative hazards for both transitions

```
coxph(Surv(time, status) ~ ccr5  + strata(failcode),
                                  data = aidssi.stack)
```

	coef	exp(coef)	se(coef)	z	p
ccr5	-0.701	0.496	0.186	-3.77	0.00016

Advantage II: assume equality

Same effect of CCR5 mutation on relative hazards for both transitions

```
coxph(Surv(time, status) ~ ccr5  + strata(failcode),
                                data = aidssi.stack)
```

	coef	exp(coef)	se(coef)	z	p
ccr5	-0.701	0.496	0.186	-3.77	0.00016

This is similar to

```
coxph(Surv(time, status!=0) ~ ccr5, data = aidssi)
```

with original data set

patnr	time	status	ccr5
14	5.05	0	WW
3	2.23	1	WW
15	10.20	1	WM
8	8.61	2	WW

Why?

$$L^{\text{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\text{AIDS}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$ `aidssi.stack`

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$ `aidssi`

$$L^{\text{SI}}(\beta) = \prod_{\substack{i=325\ldots \\ \texttt{status2}_i=1}}^{648} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\text{SI}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$ `aidssi.stack`

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=2}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$ `aidssi`

Why?

$$L^{\mathrm{AIDS}}(\beta) = \prod_{\substack{i=1\ldots \\ \texttt{status2}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\mathrm{AIDS}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi.stack}$$

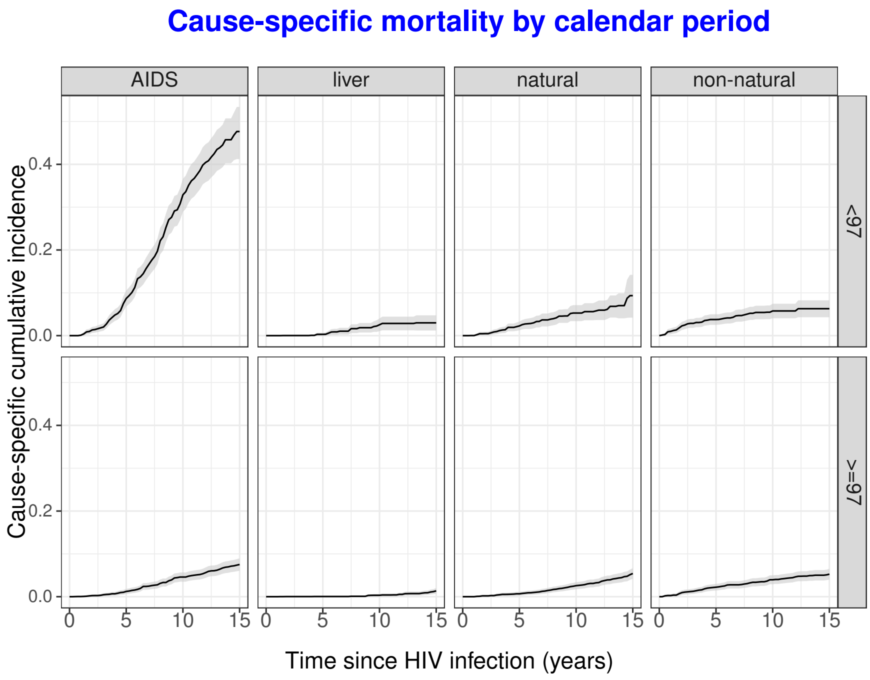
$$= \prod_{\substack{i=1 \\ \texttt{status}_i=1}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi}$$

$$L^{\mathrm{SI}}(\beta) = \prod_{\substack{i=325\ldots \\ \texttt{status2}_i=1}}^{648} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R_{\mathrm{SI}}(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi.stack}$$

$$= \prod_{\substack{i=1 \\ \texttt{status}_i=2}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right) \qquad \texttt{aidssi}$$

Same as partial likelihood based on `aidssi` data:

$$\prod_{\substack{i=1 \\ \texttt{status}_i>0}}^{324} \left(\frac{\exp(\beta \times \texttt{ccr5}_i)}{\sum_{j \in R(t_i)} \exp(\beta \times \texttt{ccr5}_j)} \right)$$



Cause-specific mortality by calendar period
AIDS
liver
natural
non-natural
<97
>=97
Cause-specific cumulative incidence
0.0
0.2
0.4
0
5
10
15
Time since HIV infection (years)

Assuming equality necessary when some event types are rare

$N = 9164$, correct for age, sex, risk group, calendar period

	MSM	idu	haemophiliac	MSW	total
alive	5052	1310	73	2252	8446
aids	158	148	65	24	397
liver	3	15	22	1	39
non natural	24	98	2	6	128
natural	68	47	14	23	154

Too few events for proper analysis per event type

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Advantage III: Proportional baseline hazards

$$\lambda_{\mathrm{SI},0}(t) = \lambda_{\mathrm{AIDS},0}(t) \times \exp(\alpha)$$

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$$\lambda_{\mathrm{SI},0}(t) = \lambda_{\mathrm{AIDS},0}(t) \times \exp(\alpha)$$

Total model now reads (AIDS: $k = 1$; SI: $k = 2$):

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{1,0}(t)\ \exp\{\alpha \times (k-1) + \beta_1 \times \mathtt{ccrAIDS}_i + \beta_2 \times \mathtt{ccrSI}_i\} \quad (1)$$

Advantage III: Proportional baseline hazards

$$\lambda_{\mathrm{SI},0}(t) = \lambda_{\mathrm{AIDS},0}(t) \times \exp(\alpha)$$

Total model now reads (AIDS: $k = 1$; SI: $k = 2$):

$$\lambda_k(t \mid \mathbf{Z}_i) = \lambda_{1,0}(t) \exp\{\alpha \times (k-1) + \beta_1 \times \texttt{ccrAIDS}_i + \beta_2 \times \texttt{ccrSI}_i\} \quad (1)$$

In R:

```
coxph(Surv(time, status) ~ ccrAIDS + ccrSI +
                    failcode, data = aidssi.stack)
```

Results:

```
              coef exp(coef) se(coef)     z       p
ccrAIDS    -1.166     0.311    0.306 -3.81 0.00014
ccrSI      -0.332     0.718    0.237 -1.40 0.16000
failcodeSI -0.184     0.832    0.148 -1.25 0.21000

Likelihood ratio test=21.5  on 3 df, p=8.12e-05  n= 648,
                                     number of events= 220
   (10 observations deleted due to missingness)
```

Outline

Observed data

$\{(v_1, x_1, e_1\delta_1), \ldots, (v_N, x_N, e_N\delta_N)\}$
Note: $t_i < c_j < v_{j'}$ in case of ties

- $x_i = \min\{t_i, c_i\}$, $\delta_i = \{t_i \leq c_i\}$, $e_i \in \{1, \ldots, K\}$
- $t_{(i)}$ ordered unique event times of any type
- $d_k(t_{(i)})$ number of events at $t_{(i)}$ of type k
- $d(t_{(i)})$ total number of events at $t_{(i)}$
- $r(t_{(i)})$ number observed at risk
- $r^*(t_{(i)})$ number “in risk set” (for subdistribution hazard)

Regression: covariables $\mathbf{Z}_i = (Z_{i1}, \ldots, Z_{iq})$

Standard survival setting: one event type

- Cox regression, **Z** time-fixed: $h(t) = h_0(t)\exp(\beta^\top \mathbf{Z}_i)$
- One-to-one relation between effects on hazard and survival

$$\overline{F}(t|\mathbf{Z}_i) = \exp\{-\int_0^t h_0(s)\exp(\beta^\top \mathbf{Z}_i)ds\} = \overline{F}_0(t)^{\exp(\beta^\top \mathbf{Z}_i)},$$

with $\overline{F}_0(t) = \exp\{-\int_0^t h_0(s)ds\} = \exp\{-H_0(t)\}$

- Estimate becomes

$$\widehat{\overline{F}}(t|\mathbf{Z}_i) = \exp\left\{-\widehat{H}_0(t)e^{\widehat{\beta}^\top \mathbf{Z}_i}\right\} = \widehat{\overline{F}}_0(t)^{\exp(\widehat{\beta}^\top \mathbf{Z}_i)}$$

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- Curves cannot cross

Outline

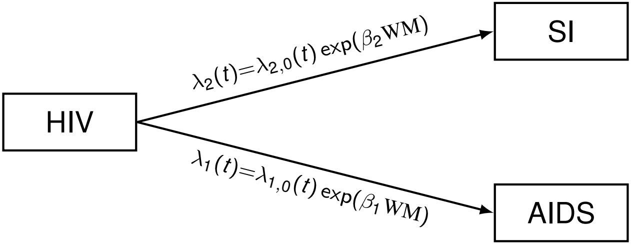
Combined analysis
- Stacked data set
- Analyses
- Standard error

Advantages combined analysis
- Test for equality
- Assume equality
- Proportional baseline hazards

From regression on hazard to cumulative scale
- Competing risks

Which Type of Analysis to Choose?
- Choices. . .
- Three types of study questions
- Etiology: marginal or competing risks analysis
- Competing risks: type of quantity

Example: effect CCR5Δ32 on AIDS and SI



Cause-specific hazard

- Cumulative progression to one end point also depends on progression to other end points
- "No one-to-one relation with cumulative scale"
- Example C: CCR5-Δ32 deletion and SI switch

$$F_{SI}(t) = \mathrm{P}\{T \leq t, E = SI\} = \int_0^t \mathrm{P}\{T \geq s\}\lambda_{SI}(s)ds$$

with $\mathrm{P}\{T \geq s\} = \exp[-\int_0^s \{\lambda_{SI}(u) + \lambda_{AIDS}(u)\}du]$

Estimate based on proportional cause-specific hazards

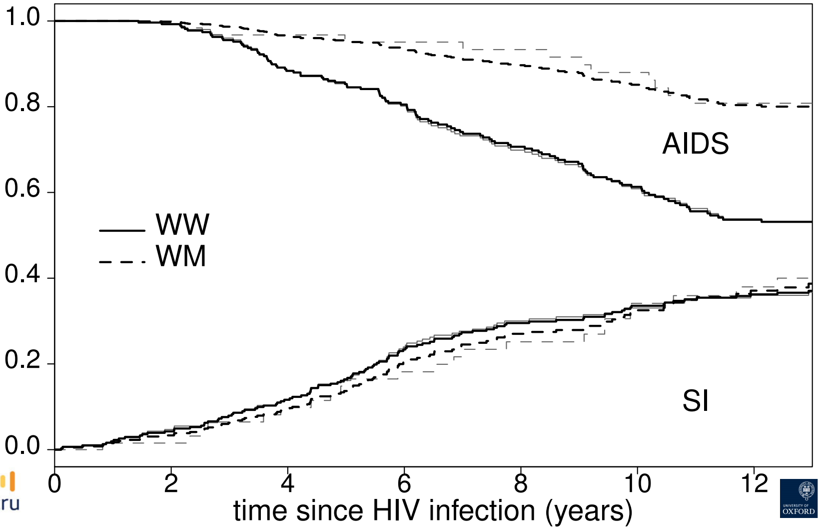
$$
\begin{aligned}
\widehat{F}_{\mathrm{SI}}(t \mid \texttt{ccr5}) &= \sum_{t_{(i)} \le t} \widehat{\lambda}_{\mathrm{SI}}(t_{(i)} \mid \texttt{ccr5}) \times \widehat{\overline{F}}(t_{(i)}- \mid \texttt{ccr5}) \\
&= \sum_{t_{(i)} \le t} \widehat{\lambda}_{0,\mathrm{SI}}(t_{(i)}) \exp\left(\widehat{\beta}_{\mathrm{SI}} \texttt{ccr5}\right) \times \\
& \exp\left\{ -\left[\widehat{\Lambda}_{\mathrm{AIDS}}(t_{(i)}- \mid \texttt{ccr5}) + \widehat{\Lambda}_{\mathrm{SI}}(t_{(i)}- \mid \texttt{ccr5}) \right] \right\}
\end{aligned}
$$

and use

$$
\begin{aligned}
\widehat{\Lambda}_{\mathrm{SI}}(t \mid \texttt{ccr5}) &= \sum_{t_{(i)} \le t} \widehat{\lambda}_{\mathrm{SI}}(t_{(i)} \mid \texttt{ccr5}) \\
&= \sum_{t_{(i)} \le t} \widehat{\lambda}_{0,\mathrm{SI}}(t_{(i)}) \exp(\widehat{\beta}_{\mathrm{SI}} \texttt{ccr5}) \\
&= \widehat{\Lambda}_{0,\mathrm{SI}}(t) \exp(\widehat{\beta}_{\mathrm{SI}} \texttt{ccr5}) \,.
\end{aligned}
$$

and similarly $\widehat{\Lambda}_{\mathrm{AIDS}}(t \mid \texttt{ccr5}) = \widehat{\Lambda}_{0,\mathrm{AIDS}}(t) \exp(\widehat{\beta}_{\mathrm{AIDS}} \texttt{ccr5})$

Cumulative incidence based on proportional cause-specific hazards model



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- Curves cannot cross
- Similar for regression model on subdistribution hazard

"Reproducing" this result: CS-RH $\neq$ cumulative effect

10 000 WW $\xrightarrow{30\%}$ 3000 **SI**

10 000 WM $\xrightarrow{26\%}$ 2600 **SI**

$\mathrm{CS} - \mathrm{RH(SI)} = 26/30 = 0.87$

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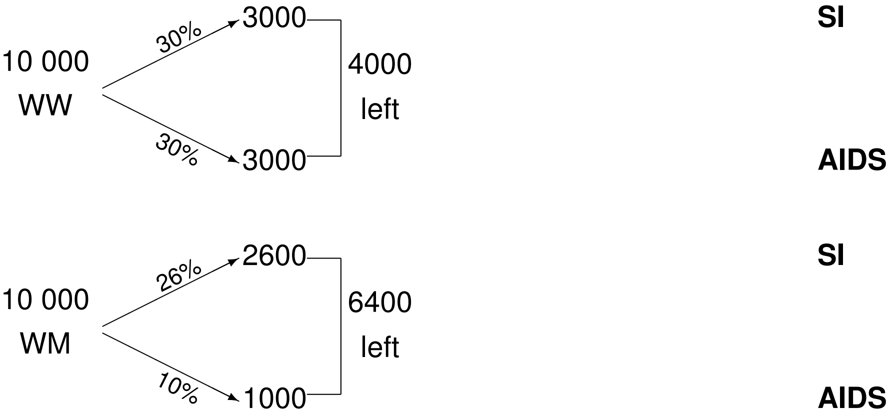

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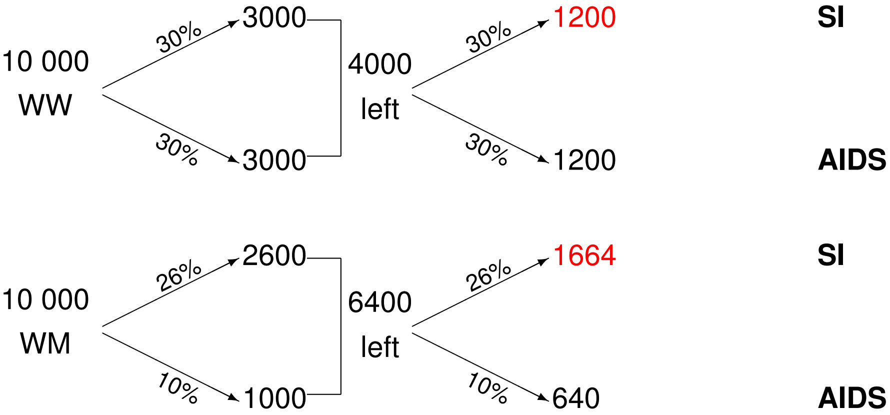

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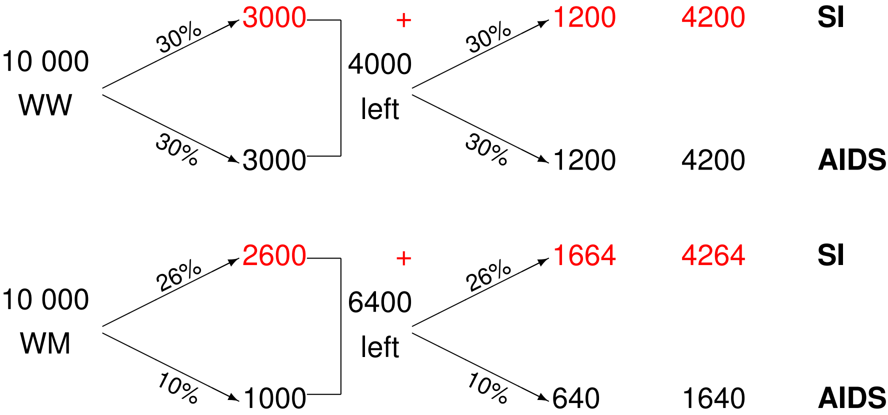

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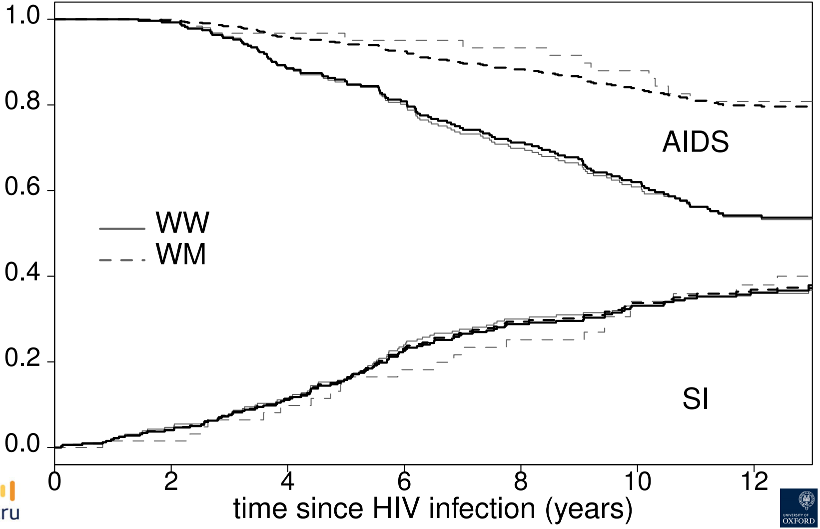
Cumulative incidence based on proportional subdistribution hazards model



Outline

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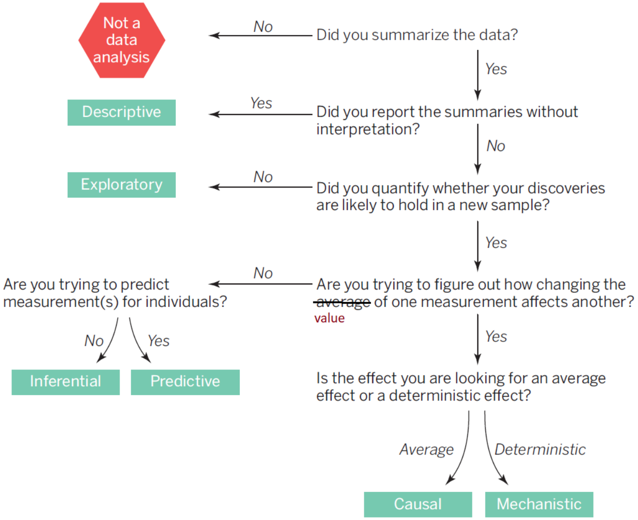
Choices...

- Interest in the outcome in the presence of competing risks or marginally?
- For competing risks analysis: cause-specific hazard or subdistribution hazard?
- For competing risks analysis: hazard or cumulative probability?

Outline

Data analysis flowchart



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- Inferential: association/correlation; risk factor analysis

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  - Cumulative event probability most relevant
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  - Validation same as in standard time-to-event setting

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  - Usually refers to real world, in which competing risks are present
  - Interventions complicate predictions
  - Validation same as in standard time-to-event setting
- Causal (see also Gillies, 2019)
  - Effect at population level, possibly by subgroup
  - Actions, interventions and manipulations
  - Biological mechanisms
  - May refer to counterfactual world without competing risks

Choices...

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Outline

Competing risk intervention or biological?

- Competing risks an intervention: marginal setting can be imagined
  - Example G: hospital discharge
  - Example H-2: death competing risk for AIDS (partly: suicide, overdose)
  - When to start intervention: what happens without intervention?

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  Treated and interpreted as censored data

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  - Example H-2: death competing risk for AIDS (partly: lung cancer or CVD unrelated diseases)

  Treated and interpreted as censored data
- Competing risk part of same mechanism: competing risks

Choices...

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- For competing risks analysis: hazard or cumulative probability?

Outline

Cause-specific or subdistribution hazard?

- Both describe competing risks setting
- Cause-specific often closer to causal effects, but any hazard is problematic
- Subdistribution directly relates to cumulative scale
  - Combination of etiology and effect on other event types. “Impact” instead of “effect”
  - Regression: parameters directly represent cumulative scale
- Some suggest to report both, especially for inferential study questions (e.g. Latouche *et al.*, 2013)

Choices...

- Interest in the outcome in the presence of competing risks or marginally?
- For competing risks analysis: cause-specific hazard or subdistribution hazard?
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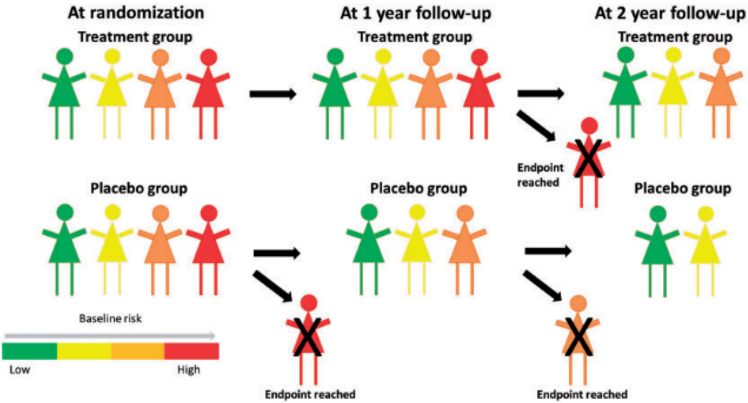
The hazard of hazard ratios

- Individuals in both arms on average equal at randomisation
- Over time
  - high risk individuals more likely to experience event
  - low risk individuals overrepresented
  - stronger selection of low risk individuals in placebo group if treatment reduces event rate
- Observed hazard ratio biased towards “no effect” (may even go in opposite direction)

Built-in selection bias



Hazard or cumulative probability?

- Prediction: cumulative probability, but may be based on regression model for the hazard
  depends on data characteristics and ease of presentation
- Etiology: value of hazard ratio does not reflect causal effect if there is unexplained variation in distribution (frailty)
- Cumulative probability: regression
  - Via cause-specific hazards, but no one-to-one relation
  - Via subdistribution hazards, one-to-one relation with cumulative scale
  - Directly via cause-specific cumulative incidence or function thereof
    In competing risks setting extra complication because cause-specific cumulative incidence neither reflects causal mechanism

An effect measure derived from cumulative risk/survival

An effect measure derived from cumulative risk/survival

- Ratio of cumulative risk
  Logistic regression at fixed τ:

$$\text{logit}[\text{P}(T \leq \tau \mid \mathbf{Z}_i)] = \beta_0 + \boldsymbol{\beta}_k^\top \mathbf{Z}_i$$

  Proportional odds logistic regression over time range:

$$\text{logit}[\text{P}(T \leq t \mid \mathbf{Z}_i)] = \log[\pi_{k,0}(t)] + \boldsymbol{\beta}_k^\top \mathbf{Z}_i$$

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  We can also use difference in cumulative risk as effect measure
- Flexible accelerated failure time model
- Difference (or ratio) of restricted mean survival time (RMST) or restricted mean time lost (RMTL)
  - Restricted: measured until fixed time point τ

$$E\{\min(T, \tau) \mid \mathbf{Z}_i\} = \beta_0 + \boldsymbol{\beta}^\top \mathbf{Z}_i$$

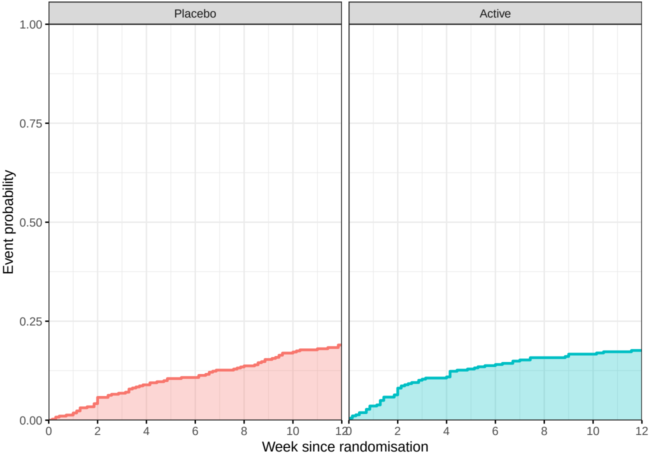
$$\log[E\{\min(T, \tau) \mid \mathbf{Z}_i\}] = \beta_0 + \boldsymbol{\beta}^\top \mathbf{Z}_i$$

Restricted mean survival time/time lost

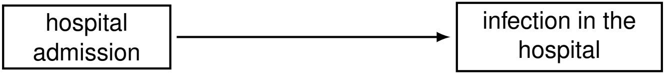
- RMST: mean event-free time up to 12 weeks

Restricted mean survival time/time lost

- RMST: mean event-free time up to 12 weeks
- RMTL: mean time lost within first 12 weeks.
  RMTL=12-RMST



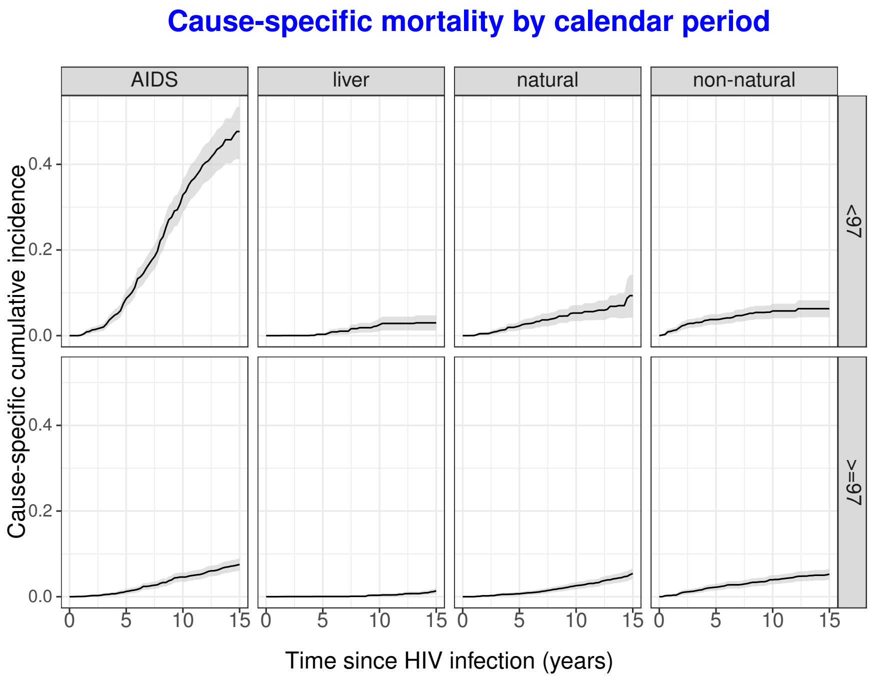
G: Staphylococcus infection during hospital stay



- Etiology (biological question): infection risk in hospital. What would happen if everyone stayed in hospital (no discharge)?
    - marginal distribution/net risk
- Predict (clinical question): burden due to infection while in hospital; discharge prevents event to occur
    - cause-specific cumulative incidence /subdistribution/crude risk
- Which of two hospitals has higher risk may depend on type of risk

The examples revisited

- Example G: staphylococcus infection in hospital
  - Etiology (what if everyone would stay in hospital): marginal hazard, Kaplan-Meier.
    Assumption: competing risk discharge independent
  - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence



Cause-specific mortality by calendar period
AIDS
liver
natural
non-natural
<97
>=97
Cause-specific cumulative incidence
0.0
0.2
0.4
0
5
10
15
Time since HIV infection (years)

The examples revisited

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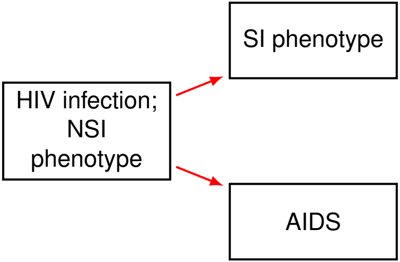
    Assumption: competing risk discharge independent
  - Prediction (how many infections are observed in hospital): cause-specific cumulative incidence
- Example B: impact of cART on causes of death.
  - Competing risks
  - Etiology: cause-specific hazard(?)
    Prediction: cause-specific cumulative incidence; subdistribution hazard for regression model

C: AIDS and SI switch after HIV infection



- SI: “syncytium inducing phenotype”. At HIV infection only NSI
- Competing risks: time to either SI switch or AIDS as first event

The examples revisited

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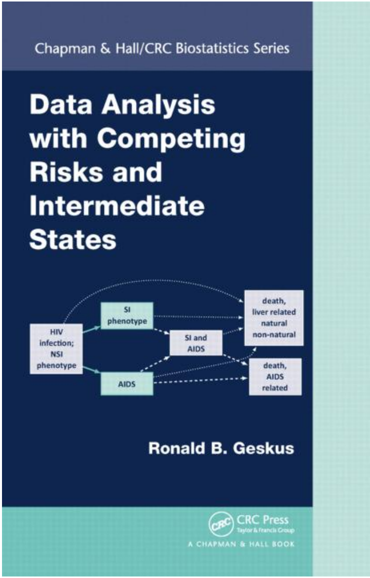
- Example H-2: natural history in IDU en MSM.
  - Marginal distribution
  - Problem: competing risks are dependent

Published by CRC Press, 2015



THANKS!

Exercise 23.1.5

Study on the effect of the use of β-blockers on prostate cancer (PCa). Individuals that used β-blockers had lower subdistribution hazard for PCa-specific death but higher for DOC.

> *To address this potential bias, we performed all analyses with the Fine and Gray competing risk regression model. In addition, we observed no increase in all-cause mortality among β-blocker users although other-cause mortality was higher, strengthening the interpretation of an association between the use of β-blockers and PCa-specific mortality.*

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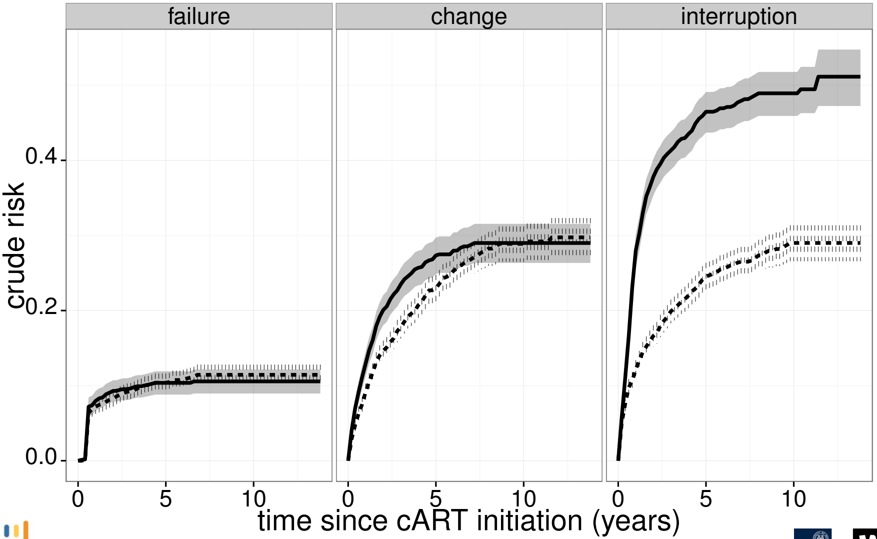
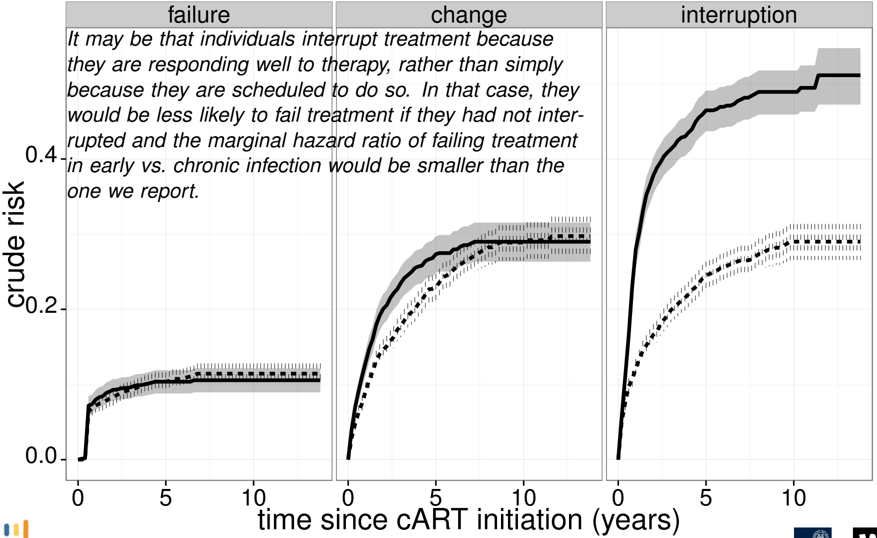
They want to study etiology, which is better described by cause-specific hazards.

H.H. Grytli *et al.*. Association between use of β-blockers and prostate cancer-specific survival. European Urology, 635-641, 2014.

Exercise 23.1.6: cART; early versus late starters



Exercise 23.1.6: cART; early versus late starters



It may be that individuals interrupt treatment because they are responding well to therapy, rather than simply because they are scheduled to do so. In that case, they would be less likely to fail treatment if they had not interrupted and the marginal hazard ratio of failing treatment in early vs. chronic infection would be smaller than the one we report.

Answer

- Assume that treatment failure and treatment interruption are the only two competing events
- Most extreme scenario: those who interrupt treatment will never fail. Marginal hazard same as the subdistribution hazard, i.e. the reported one.
- It may be true if the effect was observed for the cause-specific hazard.